

Geometry, Perimeter & Area — Level 2 — Answer Key

1. **C) 124 sq m** — Outer area = $20 \times 15 = 300$ sq m. A path 2 m wide running just inside the boundary leaves an inner rectangle of $(20-4) \times (15-4) = 16 \times 11 = 176$ sq m. Path area = $300 - 176 = 124$ sq m.
2. **C) 275 sq m** — A path built outside adds 2.5 m all around, so the outer rectangle becomes $(30+5) \times (20+5) = 35 \times 25 = 875$ sq m. Inner (the park itself) = $30 \times 20 = 600$ sq m. Path area = $875 - 600 = 275$ sq m.
3. **C) 600 sq cm** — Outer area = $40 \times 30 = 1,200$ sq cm. The photo itself (inner rectangle, 5 cm in from each side) = $(40-10) \times (30-10) = 30 \times 20 = 600$ sq cm. Frame area = $1,200 - 600 = 600$ sq cm.
4. **C) 26 sq m** — Total area = $5 \times 4 + 3 \times 2 = 20 + 6 = 26$ sq m (the two sections' areas simply add, regardless of exactly how they're joined).
5. **C) 87 sq cm** — Original area = $12 \times 8 = 96$ sq cm. Removing the 3×3 corner square removes 9 sq cm. Remaining area = $96 - 9 = 87$ sq cm.
6. **B) 2 times** — Perimeter = $2(l+b)$. Doubling both: new perimeter = $2(2l+2b) = 2 \times 2(l+b) = 2 \times$ the original perimeter.
7. **C) 9 times** — Area = side^2 . Tripling the side: new area = $(3 \times \text{side})^2 = 9 \times \text{side}^2 = 9 \times$ the original area.
8. **C) 48 cm** — Five 4 cm squares in a row form a rectangle 20 cm long (5×4) and 4 cm wide. Perimeter = $2(20+4) = 48$ cm.
9. **C) 90 sq cm** — Each tile has area $5 \times 3 = 15$ sq cm. Six tiles placed with no gaps or overlaps always cover a total area of $6 \times 15 = 90$ sq cm, however they're arranged.
10. **D) 975 sq m** — Field area = $40 \times 25 = 1,000$ sq m. Flower bed area = $5 \times 5 = 25$ sq m. Uncovered area = $1,000 - 25 = 975$ sq m.
11. **C) 32 cm** — The original rectangle's perimeter is $2(10+6) = 32$ cm. Cutting a rectangular notch out of a corner removes two short stretches of the original boundary but adds two new cut edges of exactly the same total length back in — so the perimeter of the L-shape is unchanged: still 32 cm.
12. **C) 4** — The plus-sign shape has a horizontal line of symmetry, a vertical line of symmetry, and both diagonal lines of symmetry — 4 in total.
13. **C) 70°** — The angles of a triangle always add up to 180° . $x = 180 - 50 - 60 = 70^\circ$.
14. **C) 26** — The full grid has $6 \times 5 = 30$ unit squares. Removing a 2×2 block removes 4 unit squares. Remaining area = $30 - 4 = 26$ square units.
15. **C) 42 cm** — Area 49 sq cm means each tile has side 7 cm (since $7 \times 7 = 49$). Two tiles side by side form a rectangle 14 cm by 7 cm. Perimeter = $2(14+7) = 42$ cm.

How to use this: 10+/15 is strong for a Level-2-focused set. Q11 (the corner-notch perimeter) is the trickiest idea here — it's worth re-deriving with pencil and paper: sketch the rectangle, cut the corner,

and measure each remaining side yourself rather than just trusting the shortcut.